



Queueing theory techniques and its real applications to health care systems – Outpatient visits

Peter O. Peter & R. Sivasamy

To cite this article: Peter O. Peter & R. Sivasamy (2019): Queueing theory techniques and its real applications to health care systems – Outpatient visits, International Journal of Healthcare Management, DOI: [10.1080/20479700.2019.1616890](https://doi.org/10.1080/20479700.2019.1616890)

To link to this article: <https://doi.org/10.1080/20479700.2019.1616890>



Published online: 23 May 2019.



Submit your article to this journal [↗](#)



Article views: 10



View Crossmark data [↗](#)



Queueing theory techniques and its real applications to health care systems – Outpatient visits

Peter O. Peter^a and R. Sivasamy^b

^aSchool of Mathematical Sciences, Shanghai Jiao Tong University, Shanghai, People's Republic of China; ^bDepartment of Statistics, University of Botswana, Gaborone, Botswana

ABSTRACT

Many organizations such as banks, airlines, health care systems, telecommunications companies and security departments routinely use queueing theory models to help determine capacity levels needed to experienced demands in a more efficient way. Although queueing models have been used in hospitals and other health care systems, its applications in this field has not been widely and extensively utilized. Given the perverseness of delays under health care and due to the fact that many facilities are trying to meet the increasing demands with tightly constrained resources, then queueing theory models can be very useful in identifying other opportunities for service improvement. In this article, we build a foundation into the investigation of queueing phenomenon through the review of some applicability of such techniques. Our emphasis is on intuitive understanding of queueing modelling and solution techniques that are useful in applications. We conclude this report by applying its insights and findings to real time data collected from one of the main health facilities in Botswana.

ARTICLE HISTORY

Received 21 August 2018
Accepted 2 May 2019

KEYWORDS

Queueing theory; capacity; health care systems; health facilities; resources and services

1. Introduction

In real life, waiting for a service is a common phenomenon. We always wait for services in restaurants, banks, schools, post offices, hospitals, etc. Long waiting list or time in the public health has always been a problem in most of the countries over the world. Patient flow is a very complex phenomenon because of the random nature of the arrival and service mechanisms for the patients. This often requires a systematic approach in future planning and management of facilities. Queueing theory and simulation methods are analytical techniques that are increasingly accepted as valuable tools to be used in modelling and analysing the inter-arrival and service times for patients coming to a health facility.

Service delays are mostly the result of a disparity between demand for a service and the capacity available to meet the current demand. Usually this mismatches and balances occurs due to the natural variability in the timing of demands and duration of time needed to provide service. The variability and interaction between the arrival and service processes makes the dynamics of service systems often more complex and as a result the prediction of congestion levels becomes difficult. In view of this, there is a need to apply queueing theory models to determine the capacity needed to achieve some desired levels of performance and improved service standards.

Queueing theory was first developed by the Danish Engineer Agner Krarup Erlang in 1904 to help

determine the capacity requirements for the Danish telephone systems [1]. It has since been used and applied to a wide range of service industries including scheduling of airlines, telephone call centres, computer networking and in management of health systems. Queueing theory forms part of mathematical statistics in the field of operations research that deals with the formation of queues. Its applicability has always found great use in validating complex simulator models, developing easy to implement models that help narrow alternatives in the initial stages of system design, providing insights into the importance and impact of variability in manufacturing as well as suggesting alternative ways of structuring a system and allocating tasks to people and machines.

Queueing theory is therefore concerned with the mathematical formulation and study of waiting lines that deals with modelling and the analysis of systems that provides service to random demands. A queueing model of a system will then be an abstract representation whose purpose is to isolate those factors that relates to the system's ability in meeting the service demands whose occurrences and duration are stochastic in nature. Typically, simple queueing models are specified in terms of the arrival processes, service mechanisms and the queueing discipline. The arrival process specifies the probabilistic structure the way demands of a service occur over time; the service mechanism specifies the number of servers and the

probabilistic structure on the duration of time required to serve the customer and queue discipline specifies the order in which waiting customers are selected from the queue service.

1.1. Queueing models

In specifying a queueing model, we must make some necessary assumptions about the probabilistic nature of the arrival and service processes. The most common assumption to make about the arrivals is that they follow a Poisson process. This results from the fact that the number of arrivals at any given point has a Poisson distribution. So if $N(t)$ is the number of arrivals during a time period of duration t and $N(t)$ has a Poisson distribution, then

$$N(t) = \frac{\exp^{-\lambda t} (\lambda t)^n}{n!} \quad (1)$$

where λ is called the rate and it is the expected number of arrivals per unit of time. Another way to characterize the Poisson is that the time between the consecutive arrivals known as the inter-arrival times (IA) has an exponential distribution.

So if IA denotes the inter-arrival times for a Poisson process with rate λ , then

$$P(IA \leq t) = 1 - \exp^{-\lambda t} \quad (2)$$

where $1/\lambda$ is the average time between the arrivals. An important property of the exponential distribution is that it is 'memory less'. This implies that the time of the next arrival is independent of when the last arrival occurred. It can be shown analytically that if patients arrive independently from one another, then the arrival process is also a Poisson process and it is for this reason that the Poisson is considered the most random arrival process. In determining this fact in a specific service system, it is useful to consider the following three properties:

- (a) Patients arrive one at a time.
- (b) The probability that a patient arrives at any given time is independent of arrival for the next patient.
- (c) The probability that the patient arrives at any given time is independent of the time.

The assumption of the Poisson process is sufficient when the three properties listed above accounts for a reasonable description of a system. Further tests can be performed to determine the goodness of fit. Such tests are based on the relationship of the standard deviation and the mean for the two distributions involved in the Poisson process. Since the exponential distribution is characterized by its standard deviation being equal to its mean, we can then look at the inter-arrival times and compute the ratio of the standard deviation to the mean and check if it is closer to one.

1.1.1. Characterization

A queueing system may be described as a logical phenomenon where customers arrive according to an arrival process to be serviced by a service facility according to a service process. Each service facility may contain one or more servers and it is generally assumed that each server can only service one customer at a time. If all servers are busy, then the customer is then forced to queue for a service. If a server becomes free again, then the next customer is then picked from the queue according to the rules given by the queueing discipline. During service, a customer might run through one or more stages of service, before departing from the system. In queueing theory, models are used to describe the characteristics of a queueing system. Some of the most commonly considered characteristics are discussed below:

- (a) Queue length which is the average number of customers in the queue waiting to get a service. Large queues may indicate poor server performances while small queues may indicate too much server capacity.
- (b) System length which is the average number of customers in the queue within the system, that is those waiting and those that are being serviced. Large values of this statistics implies congestion and possibly customer dissatisfaction and potential need for greater service capacity.
- (c) Waiting time in the queue explains the average time that a customer has to wait in the queue to get service. Long waiting times are directly related to customer dissatisfaction and potential loss of future revenues while very small waiting times may also indicate too much service capacity.
- (d) Total time in the system gives the average time that a customer spends in the system, from the instant of joining the queue to service completion. Large values of this statistic is indicative of the need to make adjustments in the service capacity.
- (e) Service idle time explains the relative frequency within which the service systems is idling and this idle time is directly related to cost. However, reducing idle time may have adverse effects on the other characteristics mentioned.

It is worth noting that the results obtained from various models are based on the assumption that the service system is operating under equilibrium or steady-state conditions. For various systems, the operating day begins in transient state with no customers in the system. It takes some initial time interval for enough customers to arrive such that a steady state does not mean that the system will reach a point where the number of customers remain constant.

1.1.2. Arrival processes

We denote by t_i the time at which the i th customer arrives and assume that the T 's are independent, continuous random variables projected from the random variable A . The underlying assumption that each inter-arrival times are controlled by the same random variables implies that the distributions of arrivals remains independent at different times. This is a stationary inter-arrival time assumption and it is worth to note that stationary inter-arrival times are often unrealistic but the approximation can be achieved by breaking the length of time into segments such that a negative inter-arrival time will be impossible to achieve hence; $P(A \leq c) = \int_0^c a(t) dt$ and $P(A > c) = \int_0^\infty a(t) dt$, we define $1/\lambda$ to be the mean or inter-arrival time. We also define λ to be the arrival rate. It is important to choose A such that the problem remains computationally tractable. The most common choice of A is through the exponential distribution with parameter λ and a density function $a(t) = \lambda \exp^{-\lambda}$, where $E(A) = 1/\lambda$ and $\text{Var}(A) = 1/\lambda^2$.

1.1.3. Service processes

We assume that the service times for different patients are independent random variables and that each patient's service time is controlled by a random variable having a density function $S(t)$. We let $1/\lambda$ to be the mean service time for patients. Generally, service times may not be consistent with the exponential distribution memory-less property and it is by this reason that we assume $S(t)$ is an Earlang distribution with shape parameter k and rate parameter $k\lambda$. In certain cases, the inter-arrival or service times may be modelled with zero variance and under such cases both the inter-arrival and service times are considered deterministic. As an example, if the inter-arrival times are deterministic, then each inter-arrival time will be exactly $1/\lambda$ and similarly for the service times it will also remain the same.

2. Related works summary review

Research has shown that waiting in lines or queues causes inconvenience to economic costs for both individuals and organizations as they try to minimize the total waiting cost and the cost of providing service to their customers [2]. Davis et al. [3] assert that providing ever-faster service, with the ultimate goal of having zero customer waiting time, has recently received managerial attentions for several reasons. Firstly in more developed countries where standards of living are high, time becomes more valuable as a commodity and consequently customers are less willing to wait for service. Secondly, most of the organizations have realized that the way they treat their customers today significantly impact on whether or not they will remain as loyal customers in future. Finally advances in

technology such as computers and the use of internet have provided facilities with the ability to provide faster services.

For these reasons, hospital administrators, physicians and managers are continuously working on finding the necessary means to deliver faster services with the believe that these long waiting queues will affect the after evaluation negatively. A clear understanding of the hospital inefficiencies and ways on how to improve them is very crucial for making the health care policy and budgeting decisions [4]. Chochran and Bhati [5] also argue that higher operational efficiency of any health facility is likely to control the cost of medical services and consequently provide more affordable care and improve access to the public health. Addressing the problems of queuing involve a trade-off between the costs of customers waiting time and cost of providing faster service.

Green [6] presented the queuing theory as applied in health care. The author discusses the relationship among delays, utilization and the number of M/M/s models, its assumptions and extensions and the application of the theory to determine the required number of servers. Nosek and Wilson [7] also reviewed the used queuing theory models in pharmacy applications with particular attention to improving customer satisfaction which can be achieved through prediction and reduction of waiting time as well as adjustment of staffing. Taylor [8] sought the optimal staffing for a hospital scheduling departments that handles phone calls whose intensity varies through the day. With known peak and off-peak periods the authors grouped periods that receive similar call intensity and determine the necessary staffing level which varies dynamically with the intensity of the calls overtime.

Most research on patient flow has concentrated on the Emergency Department (ED) and how to improve its flows from within. There are a few exceptions that offer a broader view. For example, Cooper et al. [9] identify a main source of ED congestions to be controlled by variability, downstream from the ED (e.g operating room schedules that are customized to Physician needs rather than being operationally optimized). In the same spirit, de Bruin et al. [10] observe that 'refused admissions at the first Cardiac Aid are primarily caused by unavailability of beds downstream the care chain'. These blocked admissions can be controlled via proper bed allocation along the rare chain of cardiac in-patients; and to support such allocations, a queueing network model was proposed with parameters that were estimated from hospital data. On a slightly different note, Sariyer et al. [11] developed the multivariate linear regression model to determine the main factors affecting Length of stay (LOS) in an Emergency Department (ED) of a large scale training hospital. The authors built and fitted a regression model for hypotheses tests with an independent group of t-tests and ANOVA under a

time sensitive setting. Broadening the view further, Hall et al. [12] develop data-based descriptions of hospital flows, starting at the highest unit-level down to the specific sub-wards (e.g. Imaging). The resulting flow charts are supplemented with descriptions of various factors that causes delays in hospitals and then some means employed to cope with such delays. There has been a growing body of research that tackles operational problems in hospitals through the application of Operations Research (OR) techniques. Kritchanhai and Hoer [13] developed a value chain model for patient flow in the OPD by classifying the facility into two types: primary and support. Their simulation results show that separating the two facilities has tremendously reduced congestion and also optimized capacity.

Green [14] surveys the potential of OR in helping reduce hospital delays with an emphasis on queueing models. Jennings and de Vericourt [15] and Green and Yankovic [16] apply queueing models to determine the number of nurses needed within a medical ward. Mandelbaum et al. [17] identified heterogeneity in Length of Stay (LOS) across Inner Wards (IWs) and this paved for fair policies of ED-to-IW routing in the sense that they balanced the workload across the wards while controlling for patient's delays. Boulton et al. [18] utilize a standard queueing model to effectively define Stroke unit capacity with less than 1% turn-away. The authors developed this queueing models from data in relation to admission rates, length of stay (LOS) and current bed availability collected from a large teaching hospital. Their study results demonstrated that the current bed capacity was considerably less than that which is required to provide effective care in maximizing potential outcome for the patients.

3. Methodology

We consider the patient's routing processes from an Outpatient Department (OPD) to the Inner Wards (IW) within a health facility. We note two main problems through out this entire process: waiting times for patients within the OPD until their transfers to inner wards are long and the allocation to the wards do not appear to be fair. This problem has motivated us to develop a model that depicts all the queueing process from OPD until a patients is assigned to an inner ward through the use of a queuing system. We employ a randomized most idle routing algorithm proposed by [19]. The algorithm appears to be fair as it chooses a server out of all idle servers at random and does not require any information on arrival rates to the system, pool sizes, or service rates at the time of routing the decision. In order to achieve additional practical insights, by accommodating some analytically intractable features and testing various routing policies, we model the OPD-to-IW process by a computer

simulation. With the simulation, we compare various routing policies, according to some fairness and performance criteria, while also accounting for the availability of information in the system. Finally, we comment on the performance of this method in relation to its practicability.

We propose the following routing policy for the queueing system and denote by $I_i(t)$ the number of idle servers in pool i at time t . We denote by $\varepsilon_i(t)$ the number of empty places in the buffer of pool i at time t and assume work conservation, that is, if $I_i(t) > 0$ then $\varepsilon_i(t) = b_i$ or equivalently if $\varepsilon_i(t) < b_i$ then $I_i(t) = 0$. By $V_i(t) = I_i(t) + \varepsilon_i(t)$ we denote the total number of vacant places (idle servers + empty places in the queue) in pool i at time t . A customer arriving at time t is routed to pool i with probability $V_i(t)/\sum_{j=1}^K V_j(t)$ unless $V_i(t) = 0$, $\forall i = 1, \dots, K$ in which case the customer leaves or joins the centralized queue depending on the system. We take $b_i = 0 (C_i = N_i)$, $\forall i \in \{1, 2, \dots, k\}$.

We model the system under randomized most idle routing process as Markov chain in continuous time $Z = \{Z_t, t \geq 0\}$ and characterize each state with the centralized queue being empty, as a K -dimensional vector $z = (z_1, z_2, \dots, z_K)$, where z_i is the total number of customers in pool i both in service and in the queue ($z_i \in \{0, 1, \dots, C_i\}$). We also define $m_z = \sum_{i=1}^K z_i$ as the total number of customers at state z . The state $(C_1, C_2, \dots, C_K)$, where all the distributed queues are full, is denoted by (C) . States in which the number of customers in the system exceeds C by l customers, are denoted by $(C + l)$, where l is the length of the centralized queue.

The whole queue connection process is modelled by a general Markovian Birth and Death process and each state $C + l$ has a birth rate λ_l and death rate μ_l and the maximum number of customers in the centralized queue is $c > 0$. These processes are connected via state C and the resulting process is reversible and its stationary distribution is calculated from the individual processes stationary distributions given by

$$\pi_z = \begin{cases} \pi_0 \prod_{i=1}^K \left(\frac{C_i!}{(C_i - z_i)! (N_i \wedge z_i)! \mu_i^{z_i}} \right) \frac{\lambda^{m_z} (C - m_z)!}{C!}, & l = 0; \\ \pi_0 \prod_{i=1}^K \left(\frac{C_i! N_i^{-b_i}}{N_i! \mu_i^{C_i}} \right) \frac{\lambda^C \prod_{i=0}^{l-1} \lambda_i}{C! \prod_{i=1}^l \mu_i}, & l > 0. \end{cases} \quad (3)$$

Here π_0 is deduced from $\sum \pi_z = 1$ and this is given by

$$\pi_0 = \left[\sum_{z_i=0}^{C_i} \dots \sum_{z_K=0}^{C_K} \left(\prod_{i=1}^K \left(\frac{C_i!}{(C_i - z_i)! (N_i \wedge z_i)! \mu_i^{z_i}} \right) \right) \frac{\lambda^{m_z} (C - m_z)!}{C!} + \prod_{i=1}^K \left(\frac{C_i! N_i^{-b_i}}{N_i! \mu_i^{C_i}} \right) \frac{\lambda^C}{C!} \sum_{l=1}^c \frac{\prod_{i=0}^{l-1} \lambda_i}{\prod_{i=1}^l \mu_i} \right]^{-1} \quad (4)$$

Note that by substituting $b_i = 0 (C_i = N_i)$, we obtain the stationary distribution under the randomized process and

on the other side, taking $b_i \rightarrow \infty, \forall i \in \{1, 2, \dots, K\}$ (keeping N_i fixed), we converge to the stationary distribution, where customers are routed to one of the K pools at random with probability $1/K$.

3.1. The general structure

Consider the queueing system projected under Figure 1 where homogeneous customers arrive according to a Poisson process with rate $\lambda > 0$, N heterogeneous servers operate independently in parallel, service times for server i are *i.i.d* exponential with rate μ_i and the system has one waiting line with infinite capacity. Each customer requires a service from any of the servers, and each server can serve only one customer at a time. The queueing discipline is FCFS (first-come-first-served), non-preemptive (the service of a customer cannot be interrupted once started), and work-conserving (there are no idle servers whenever there are delayed customers in the queue). In addition, we assume that all inter-arrival and service times are statistically independent.

3.2. Simulation analysis

A generic computer simulation program which modelled the OPD-to-IW processes was built within the R software. The simulation parameters were estimated from the empirical quantitative data obtained from the hospital records and its results were validated against real data. Our aim was also to find which routing algorithm seems optimal in terms of performance and fairness. We define various fairness and performance measures to form a single integrated quality criterion.

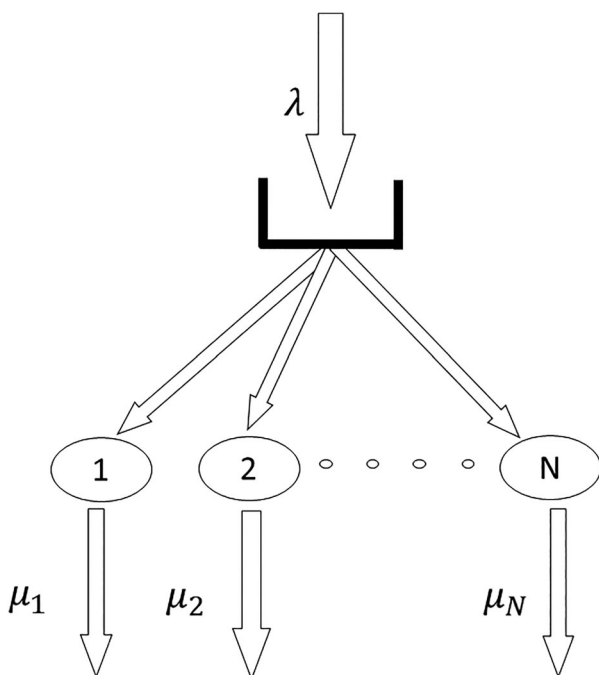


Figure 1. Queueing system with heterogeneous servers.

We considered several routing algorithms while accounting for availability of information in the system and implemented them in to the simulation process for optimal comparison with each other.

3.3. Simulation model

Through out our simulation process we use the structure as depicted in Figure 2 as an underlying simulation model where we propose four heterogeneous server pools representing the four inner wards A–D, and each pool contains a number of *i.i.d* servers corresponding to the number of beds in each ward. Of great importance, we attempted to account for the fact that patients waiting times in the OPD is not only caused by lack of bed in the designated ward as is normally the case in classical queueing systems. We thus divided the waiting times into two phases, delay when there is an available bed and waiting when all the beds are occupied. The boarding time (that time which the staff requires for preparation) was modelled as an exponential random variable with the mean depending linearly on the occupancy rate of each ward at the time of allocation. Our simulation model however has some limitations regarding a single customer/patient class and we also assumed that each pool has a dedicated queue of infinite capacity although the finite-queue model is more appropriate.

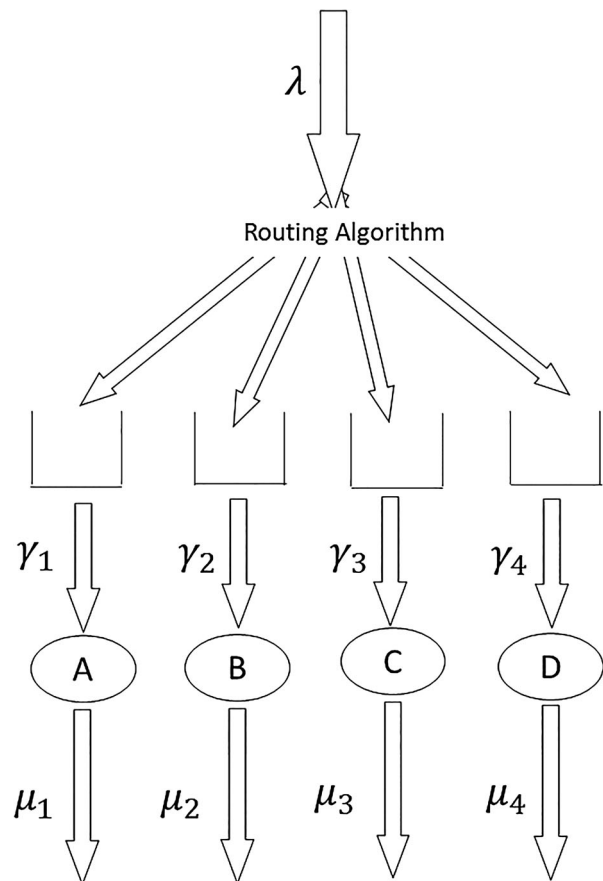


Figure 2. Simulation model.

3.4. Measuring performance of a queueing system

Queueing system can be studied in so many ways to investigate the rate at which it is performing and determine the quality of the service. Generally this approach depends on the followings:

- (a) The rate at which customers arrive
- (b) The time taken to serve each customer
- (c) The number of servers available.

The performance of the queue can also be measured by:

- (a) The time customers spend waiting
- (b) The average number of customers in the waiting line
- (c) The utilization rate of the server
- (d) Cost of operating the system

In this section, we will compute several steady-performance measures for the queueing model under the randomized routing algorithm. We analyse the system with a general queue structure and its stationary distribution. We denote the stationary queue length (number of customers in the queue in steady state) by L_q , and the stationary waiting time in the queue by W_q . Firstly we find $P(W_q > 0)$ as the steady-state probability that all servers are busy or the steady-state probability that a customer arriving to the system is delayed:

$$\begin{aligned} P(W_q > 0) &= \sum_{l=0}^c \pi_{N+l} \\ &= \pi_0 \frac{\lambda^N}{N! \prod_{i=1}^N \mu_i^{N_i}} \sum_{l=0}^c \frac{\prod_{i=0}^{l-1} \lambda_i}{\prod_{i=1}^l \mu_i} \end{aligned} \quad (5)$$

where π_0 is given in (4) above and we can obtain the queue length distribution as

$$\begin{aligned} P(L_q = l | W_q > 0) &= \frac{\pi_{N+l}}{\sum_{i=0}^c \pi_{N+i}} \\ &= \frac{\frac{\prod_{i=0}^{l-1} \lambda_i}{\prod_{i=1}^l \mu_i}}{\sum_{k=0}^c \frac{\prod_{i=0}^{k-1} \lambda_i}{\prod_{i=1}^k \mu_i}}, \end{aligned} \quad (6)$$

$\forall l = 0, 1, \dots, c.$

$$\begin{aligned} P(L_q = l) &= P(L_q = l | W_q > 0) P(W_q > 0) \\ &= \frac{\frac{\prod_{i=0}^{l-1} \lambda_i}{\prod_{i=1}^l \mu_i}}{\sum_{k=0}^c \frac{\prod_{i=0}^{k-1} \lambda_i}{\prod_{i=1}^k \mu_i}} P(W_q > 0), \end{aligned} \quad (7)$$

$\forall l = 1, 2, \dots, c.$

The performance measure of most interest in the loss model is the steady-state loss probability, which is the probability that the system is in state (N) where all the servers are busy. By Poisson arrivals, this is given by the probability that an arriving customer is blocked and we find the following expression for the loss probability:

$$\begin{aligned} P(\text{block}) &= \pi_N = \pi_0 \frac{\lambda^N}{N! \prod_{i=1}^K \mu_i^{N_i}} \\ &= \frac{\frac{\lambda^N}{N! \prod_{i=1}^K \mu_i^{N_i}}}{\sum_{y_1=0}^{N_1} \dots \sum_{y_N=0}^{N_N} \left(\prod_{i=1}^K \binom{N_i}{y_i} \right) \frac{(N - m_y)!}{N!} \frac{\lambda^{m_y}}{\prod_{i=1}^K \mu_i^{y_i}}} \end{aligned} \quad (8)$$

Another popular queue structure is the Delay model with infinite queue and the transition rates of the Birth and Death queue processes are $\lambda_l = \lambda, \forall l = 0, 1, 2, \dots$ and $\mu_l = \sum_{i=1}^K N_i \mu_i, \forall l = 1, 2, \dots$. We note that this queue process is similar to the one formed by $M/M/N$ with arrival rate λ and individual service rate $\mu = \sum_{i=1}^K N_i \mu_i / N$ (the average service rate of the system servers). Hence performance measures concerned with delay (queue length and waiting time distributions) resemble the corresponding measures of this $M/M/N$ system given a steady-state delay.

For delay models with infinite queues, we denote the total traffic intensity of the system by

$$\rho := \frac{\lambda}{\sum_{i=1}^K N_i \mu_i} \quad (9)$$

We assume that $\rho < 1$ and substituting the transition rates of the queue process in (5) and taking $c \rightarrow \infty$, we obtain the steady-state delay probability:

$$\begin{aligned} P(W_q > 0) &= \pi_0 \frac{\lambda^N}{N! \prod_{i=1}^N \mu_i^{N_i}} \sum_{l=0}^{\infty} \frac{\lambda^l}{(\sum_{i=1}^K N_i \mu_i)^l} \\ &= \pi_0 \frac{\lambda^N}{N! \prod_{i=1}^N \mu_i^{N_i}} \left(\frac{1}{1 - \rho} \right) \end{aligned} \quad (10)$$

where π_0 is obtained from (4) and is given by

$$\begin{aligned} \pi_0 &= \left[\sum_{y_1=0}^{N_1} \sum_{y_2=0}^{N_2} \dots \sum_{y_K=0}^{N_K} \left(\prod_{i=1}^K \binom{N_i}{y_i} \right) \right. \\ &\quad \times \left. \frac{(N - m_y)!}{N!} \frac{\lambda^{m_y}}{\prod_{i=1}^K \mu_i^{y_i}} \left(\frac{1}{1 - \rho} \right)^{(m_y - N + 1)} \right]^{-1} \end{aligned} \quad (11)$$

Finally, we deduce the queue length and waiting time distributions by substituting the transition rates (6) and (7), and taking $c \rightarrow \infty$. The performance measures obtained are indeed similar to the measures of the corresponding $M/M/N$ system:

$$\begin{aligned} P(L_q = l | W_q > 0) &= \rho^l (1 - \rho), \\ \forall l \in \{0, 1, 2, \dots\} \end{aligned} \quad (12)$$

$$P(L_q = l) = P(W_q > 0) = \rho^l(1 - \rho)P(W_q > 0),$$

$$\forall \in \{1, 2, \dots\}$$

$$EL_q = E(L_q | W_q > 0)P(W_q > 0) = \frac{\rho}{1 - \rho}P(W_q > 0)$$

$$EW_q = \frac{EL_q}{\lambda} = \frac{P(W_q > 0)}{\sum_{i=1}^K N_i \mu_i (1 - \rho)}$$

3.5. Application to real hospital data

This section outlines the layout of one health facility in Botswana. A hospital is an integral part of the society function of which is to provide medical care and services to people. The hospital of interest was opened in 1995 to provide medical services to members of the army and their families. It is currently housing different departments and wards which includes; Laboratory sections (Haematology, Chemical and microbiological labs, Serology, Blood banks, etc.), Maternal and child health clinic, Eye clinic, Pharmacy, Physio, X-ray department, Theatre and outpatient department (OPD).

The medical Outpatient department (OPD) is the busiest of all the departments and is the point of contact where records on arrival, waiting and service times for patients were obtained. It consists of the Minor surgery section, Accident and emergency department, Dressing and injection rooms, Orthopaedic clinic, Paediatric health services, General consultation clinic and Inner wards (IW). Under the General consultation clinic, patients are firstly registered at the registration counter upon arrival before proceeding to the next section where they are subjected to routine medical check-ups and tests. Patients are registered and assisted in order of their arrival timings through the registration counter and the medical check-up stage until attended to by a Physician. The capacity layout of these Inner Wards is projected under Table A4. Ward A has the capacity to admit more patients compared to the rest with an average Length of Stay (LOS) of 5 days. This makes it the biggest ward even in terms of standard capacity followed by Ward B.

The admission of patients to one of the wards, A–D is done on the basis of an allocation algorithm which generally promotes fairness and balanced load across the wards. This algorithm accepts a patient's category as an input parameter and returns a ward for the patient as an output. In the past, patient's allocation was decided according to a fixed 'table of duty' for the wards (in the sense that everyday, the ward on duty had to accommodate all the incoming patients), but allocation was subject to ward's approval. Each ward had the authority to refuse admitting the patient

and consequently, waiting times to the wards were extremely long. Most recently, more patient's routing policies have been proposed to determine fair placement and admission of patient's into wards, see [19].

Data on the flow patterns of patients has been recorded under three settings (Day 1, Day 2 and Day 3) selected at random under different timings. This data collection was also extended to other remaining Quarters of the year in a similar fashion using past hospital records. The adopted approach allows us to make a comparison across the three Quarters within the same data collection settings. We employ the R software for the computation of all the estimates and steady states necessary to establish model operating characteristics and use this results to validate the simulated model given under Section 3.3. Results for all the three Quarters are projected in the [Appendix](#) page under [Tables A1–A3](#) respectively.

4. Discussions

The results from all the datasets collected under three Quarters in all the three settings indicate that Day 2 is generally the busiest day for the first and second Quarters as verified by a queue utilization rate (P) of 82 and 79% respectively. The third Quarter has recorded the utilization rate of 80% under Day 3 which differs significantly to both first and second Quarters. On average, we also note that a patient would be expected to spend at least 8–10 minutes across all the three Quarters in the system to complete the entire service path and this was a fair standard looking at the arrival and service rates of patients for this particular hospital. Generally the mean arrival rates for patients under all the three settings is much less than the mean service rates and this accounts for a reasonable rate in the utilization factor of the server except under Day 2, see summaries of these results as projected in [Tables A1–A3](#) under the [Appendix](#) section. We also point out that while the focus is on this particular hospital of interest, the average expected time of 10 minutes may not be generalized to reflect on the same conclusion across other hospitals. However, we do acknowledge that each case treated separately has its own merits and demerits hence the application of this methodology will become handy in establishing better insights for Hospital Managements in relation to their specific situations.

5. Conclusion

Generally health care systems are major determinants of human quality of life. Queueing theory is therefore a very powerful tool that is often overlooked especially under hospital environments for performance management. Proper application of this effective management tool can yield impressive results. There are several

factors relating to physical, psychological and emotional feelings that affects patient's perception on the waiting experience. Through better understanding of the techniques of Queueing theory and various measures associated with patients waiting time, then hospital managers can make decisions that have a beneficial impact on the satisfaction of all the stake holders. In reality, waiting in lines will always be prevalent within every society including hospital environments and as the health industry continues to grow, the application of Queueing theory can be used to develop and identify other opportunities for service improvement. Boulton et al. [18] argue that Queueing theory provides an additional framework to help clinicians and managers more effectively in future plans. The authors highly recommend that such modelling techniques be implemented for wider use in health care and planning. We therefore conclude that although actually reducing wait times may not always be possible (due to lack of resources or staffing limitations), it remains possible to improve some patient-centred aspects of care services, which can mitigate the patients' dissatisfaction with waiting periods and enable more positive perceptions regarding the services they receive.

Acknowledgments

The authors would like to thank their colleagues and peer reviewers for their valuable inputs, comments and suggestions.

Disclosure statement

No potential conflict of interest was reported by the authors.

Notes on contributors

Peter O. Peter, BA (Statistics), MA (Statistics) PhD Student: Peter O. Peter is currently a Statistics PhD Student under School of Mathematical Sciences at Shanghai Jiao Tong University. His main Area of Research is on Data Science with more focus relating to High Dimension Statistical Inference, Data Mining and Machine Learning. Their current Project in on Robust Predictors Ranking Methods for Variable and Model Selection via Extended Bayesian Information Criterion under Ultra High-dimensional Setting ($p > n$).

R. Sivasamy, PhD is a Professor of Statistics and has taught both undergraduate and graduate statistics courses at the University of Botswana. His main area of research specialization is on stochastic processes and Markov chains. He has published extensively within the field of Quality Control, Clinical trials, Queueing problems and Inventory Management.

References

- [1] Earlang AK. The life and the works of A. K. Earlang. Copenhagen, Denmark: Copenhagen Telephone Co, 1948.; 1948.
- [2] Kartz KL. Prescription for the waiting-in-line blues: entertain, enlighten, and engage. *Sloan Manage Rev.* 1991;32(2):44.
- [3] Davis MM, Aquilano JN, Chase BR. Fundamentals of operations management. Vol. 4. State College (Pennsylvania): Irwin McGraw-Hill; 2003.
- [4] Wilson MJ, Nguyen K. Bursting at the seams: improving patient flow to help America's emergency departments. Washington, DC: Urgent Matters, The George Washington University Medical Center, School of Public Health and Health Services; 2004.
- [5] Chochran KJ, Bhati A. Multi-stage stochastic methodology for whole hospital bed planning under peak loading. *Int J Ind Syst Eng.* 2006;1(1/2):8–35.
- [6] Green L. Queueing analysis in healthcare. In: *Patient flow: reducing delay in health care delivery*. New York, Boston, MA: Springer; 2006. p. 281–308.
- [7] Nosek AR, Wilson PJ. Queueing theory and customer satisfaction: a review of terminology, trends and applications to pharmacy practice. *Hosp Pharm.* 2001;3(36):275–279.
- [8] Taylor S. Waiting for service: the relationship between delays and evaluations of service. *J Mark.* 1994;58(2):56–69.
- [9] Cooper AB, Litvak E, Long MC, et al. Emergency department diversion: causes and solutions. *Acad Emerg Med.* 2001;8:1108–1110.
- [10] de Bruin AM, van Rossum AC, Visser MC, et al. Modeling the emergency cardiac in-patient flow: an application of queueing theory. *Health Care Manag Sci.* 2007;10:125–137.
- [11] Sariyer G, Ataman MG, Kiziloglu I. Factors affecting length of stay in the emergency department: a research from an operational viewpoint. *Int J Healthc Manag.* 2018;1–10. Available from: <https://doi.org/10.1080/20479700.2018.1489992>.
- [12] Hall R, Belson D, Murali P. Modeling patient flows through the healthcare system. In: Hall RW, editor. *Patient flow: reducing delay in healthcare delivery*. Vol. 1. Boston, MA: Springer 2006. p. 1–45.
- [13] Kritchanai D, Hoeur S. Simulation modeling for facility allocation of outpatient department. *Int J Healthc Manag.* 2018;11(3):193–201. Available from: <https://doi.org/10.1080/20479700.2017.1359920>.
- [14] Green LV. Identifying good nursing levels: a queueing approach. *Tutor Oper Res.* 2008;1:1–16.
- [15] Jennings OB, de Vericourt F. Dimensioning large-scale membership services. *Oper Res.* 2008;56:173–187.
- [16] Green L, Yankovic N. Identifying good nursing levels: a queueing approach. *Oper Res.* 2011;59:942–955.
- [17] Mandelbaum A, Momcilovic P, Tseytlin Y. Queueing systems with heterogeneous servers: on fair routing of patients in emergency departments. *Manage Sci.* 2011;1(1):65–75.
- [18] Boulton J, Akhtar N, Shuaib A, et al. Waiting for a stroke bed: planning stroke unit capacity using queueing theory. *Int J Healthc Manag.* 2016;9(1):4–10. Available from: <https://doi.org/10.1080/20479700.2015.1101910>.
- [19] Tseytlin Y. Queueing systems with heterogeneous servers: on fair routing of patients in emergency departments. Technion-Israel Institute of Technology, Faculty of Industrial Engineering and Management; 2009.

Appendix. Results

Table A1. First Quarter performance results (mins = minutes).

Day	Queue performance results					
	P	P_0	L_0	L_q	W_s	W_q
Day 1	0.75	0.25	3 patients	2 patients	6 mins	4 mins
Day 2	0.82	0.18	5 patients	4 patients	10 mins	5 mins
Day 3	0.67	0.33	2 patients	1 patient	5 mins	2 mins

Table A2. Second Quarter performance results (mins = minutes).

Day	Queue performance results					
	P	P_0	L_0	L_q	W_s	W_q
Day 1	0.72	0.28	2 patients	3 patients	7 mins	5 mins
Day 2	0.79	0.21	6 patients	4 patients	9 mins	7 mins
Day 3	0.63	0.37	3 patients	2 patient	6 mins	3 mins

Table A3. Third Quarter performance results (mins = minutes).

Day	Queue performance results					
	P	P_0	L_0	L_q	W_s	W_q
Day 1	0.77	0.23	4 patients	3 patients	5 mins	4 mins
Day 2	0.80	0.20	5 patients	5 patients	8 mins	6 mins
Day 3	0.60	0.40	3 patients	2 patient	4 mins	3 mins

Table A4. Internal wards capacities.

	Ward A	Ward B	Ward C	Ward D
standard capacity (number of beds)	30	25	20	15
maximal capacity (number of beds)	35	30	28	22
Average Length of Stay (LoS)	5	4	6	4